

WORKING TITLE

by

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Abstract

My thesis is about bla bla

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Chapter 1

Outline

1. Background/Preliminaries

(a) (Maybe) Virtual double categories

(b) Multicategories/operads

i. Definition and examples

ii. Operads difference

(c) ∞ -cosmos

i. Definition and example using simplicial sets

ii.

(d) Dendroidal Sets

2. Main Contents

(a) (Maybe) some of the work on lax monoidal structures, but using the virtual double categorical setting.

(b) Showing the model of infinity operads follows most of the axioms of an ∞ -cosmos.

- (c) Introduce the definition of weak ∞ –cosmos and show similar results from preliminaries still follows

Chapter 2

Background

2.1 Multicategories

EXPLAIN THE SOURCE OF THIS MATERIAL. “Multicategories were first defined in ...” “Our exposition follows ...” yes Multicategories are a generalization of categories. In categories, arrows have a single source/domain to a single target/codomain, in a multicategories the arrows have a finite sequence of sources/inputs to a single target/output.

Definition 2.1.1. A *multicategory* $\mathcal{M}$ consist of a set S of objects and for each $n \geq 0$ and each sequence $s_1, \dots, s_n, t$ of objects a set

$$\mathcal{M}(s_1, \dots, s_n; t)$$

of multiarrows/operations, thought of as taking n inputs of objects $s_1, \dots, s_n$ respectively to the object output t . Moreover, there are three kinds of structure maps:

- For each object s , an identity/unit $1_s \in \mathcal{M}(s; s)$

- For any sequence of objects $s_1, \dots, s_n, t$ and any n -tuple of sequences $d_1^i, \dots, d_{k_i}^i$ of objects for $i = 1, \dots, n$, a composition function of multiarrows/operations

$$\gamma : \mathcal{M}(s_1, \dots, s_n; t) \times \prod_{i=1}^n \mathcal{M}(d_1^i, \dots, d_{k_i}^i; s_i) \rightarrow \mathcal{M}(d_1^1, \dots, d_{k_1}^1, d_1^2, \dots, d_{k_n}^n; t),$$

usually written $\gamma(p, q_1, \dots, q_n) = p \circ (q_1, \dots, q_n)$ or even just $p(q_1, \dots, q_n)$

Furthermore these structure maps have to satisfy a number of axioms, which express that composition is unital and associative (Heuts and Moerdijk, 2022)

Remark 2.1.2. Let $\mathcal{M}$ be a multicategory, $f \in \mathcal{M}(s_1, \dots, s_n; t)$ and $g \in \mathcal{M}(d_1, \dots, d_l; s_i)$ for some $1 \leq i \leq n$. It will be convenient to have the following notation:

$$f \circ_i g := f(1_{s_1}, \dots, g, \dots, 1_{s_n})$$

Thus, $f \circ_i g$ is the composition of f with g in the i th variable.

Example 2.1.3. You can think of the multiarrows in $\mathcal{M}(s_1, \dots, s_n; t)$ as operations, taking inputs of types $s_1, \dots, s_n$ respectively to an output of type t . **FIX A SINGLE SET X ? and then n -ary multiarrows are operations on X .** For example, for a fixed family of sets $X_{c \in C}$, define a multicategory $\mathcal{M}$ whose objects is the set of colors C , and whose multiarrows are defined by $\mathcal{M}(c_1, \dots, c_n; c)$ to be the set of functions

$$p : X_{c_1} \times \dots \times X_{c_n} \rightarrow X_c$$

Composition in $\mathcal{M}$ is defined as composition of functions.

Example 2.1.4. If $\mathcal{C}$ is a symmetric monoidal category and C is a set of objects of $\mathcal{C}$, then one obtains a multicategory with this set of objects by defining

$\mathcal{M}(c_1, \dots, c_m; t)$ to be the set of morphisms $c_1 \otimes \dots \otimes c_m \rightarrow t$ in $\mathcal{C}$. We will sometimes write $\mathcal{C}^{\otimes}$ for the multicategory obtained in this way, when $\mathcal{C}$ is the set of all objects of $\mathcal{C}$ (granted it is small).

Example 2.1.5. A (small) category $\mathcal{C}$ can be considered as a multicategory $\mathcal{C}_m$ which only have unary arrows. I.e. the set $\mathcal{C}_m(c_1, \dots, c_n; c)$ is always empty unless $n = 1$.

We can observe that multicategories form a category $\mathcal{MCat}$.

Definition 2.1.6. For two multicategories $\mathcal{M}$ and $\mathcal{N}$, a *multifunctor* $\phi : \mathcal{M} \rightarrow \mathcal{N}$ consist of:

- A function $f : O_{\mathcal{M}} \rightarrow O_{\mathcal{N}}$ between the corresponding sets of objects
- For each sequence $s_1, \dots, s_n, t$ of objects of $\mathcal{M}$, a map

$$\phi_{s_1, \dots, s_n, t} : \mathcal{M}(s_1, \dots, s_n; t) \rightarrow \mathcal{N}(f(s_1), \dots, f(s_n); f(t)).$$

These maps being compatible with compositions and units in the obvious way.

2.1.1 Trees

Definition 2.1.7. A *tree* T consist of a finite set V of *vertices*, a nonempty finite set E of *edges*, a distinguished element $r \in E$ called the *root*, together with the following data:

- (1) a function $I : E - \{r\} \rightarrow V$, which we think of as assigning to an edge e the vertex $I(e)$ of which is an input.

(2) A function $O : V \rightarrow E$, assigning to each vertex v its output edge $O(v)$

For each edge e other than the root r , we obtain a sequence of edges starting at e by repeatedly applying $O \circ I$. We demand that this sequence ends in the root r after finitely many steps, for an arbitrary starting edge. The edges in the complement of the image of O are called *leaves* of the tree T . The vertices in the complement of the image of I are called *stumps*, or *nullary vertices*. An *outer edge* is an edge that is either the root or a leaf. An *inner edge* is any other edge of T . Such an edge is both an output edge and an input edge for some other vertex.

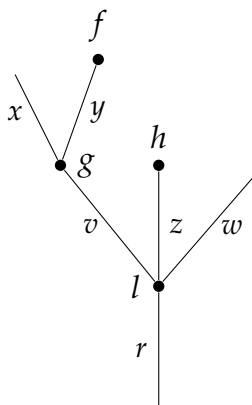
When writing T for a tree, we will usually write $E(T)$ and $V(T)$ for its sets of edges and vertices, respectively.

Example 2.1.8. The smallest possible tree consists of a single edge and no vertices; this tree will be denoted by μ

Example 2.1.9. The next smallest tree consists of a single edge and single vertex, pictured as:



Example 2.1.10. A bigger and more typical tree T with $E(T) = \{x, y, z, w, v, r\}$ and $V(T) = \{f, g, h, l\}$:



Bibliography

Heuts, Gijs and Ieke Moerdijk (2022). *Simplicial and Dendroidal Homotopy Theory*. Springer Cham.